UNIVERSITY OF HAWAIʻI AT MĀNOA

# CAPPY User Guide

*Capture and Acquisition Program in Python*

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**Version:** 1.0

**Date:** March 19, 2026

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# 1 Introduction

CAPPY, short for **Capture and Acquisition Program in Python**, is a waveform acquisition and archive system for AlazarTech digitizers using the `atsapi` interface. It is designed for triggered capture, live monitoring, and structured archival of reduced data and selected waveform snippets.

This guide serves as an operating reference for acquisition sessions and archive review.

In this guide, **CAPPY** refers to the acquisition program and **CAPPY.ARCH** refers to the archive browser.

# 2 System overview

CAPPY is designed to support:

1. triggered pulse acquisition,
2. reduced-data and waveform storage,
3. live monitoring and archive inspection.

The program combines hardware control, DMA-based acquisition, numerical reduction, selective waveform storage, and archive browsing in one workflow.

# 3 Program functionality

CAPPY v1.3 and CAPPY v1.0 are Python-based acquisition and archiving systems for AlazarTech digitizers through the `atsapi` interface. The main difference between the two versions is board identity and calibration. The acquisition workflow remains the same.

At startup, the program loads a YAML configuration containing acquisition, channel, trigger, waveform-save, runtime, archive, and preview settings. These values are validated before hardware setup.

The program then configures the digitizer clock, active channels, voltage range, coupling, impedance, trigger source, trigger slope, and record length. Acquisition is performed as triggered records with pre-trigger and post-trigger regions. Multiple records are grouped into DMA buffers for transfer to host memory.

After each completed buffer, the raw payload is interpreted using the active channels, samples per record, records per buffer, and ADC resolution. The data are reshaped into waveform arrays for Channel A and, when enabled, Channel B.

ADC codes are converted to calibrated voltages. Each record is then reduced into compact observables such as baseline, integrated area, and peak amplitude.

Each record is also assigned indexing and timing metadata, including DMA buffer number, record number within buffer, global record number, and timestamp. These fields are used later by the archive browser.

Waveforms are saved according to selection rules. The program can save every $N$th waveform, waveforms above a peak threshold, waveforms above an area threshold, or combinations of these rules. A per-second save cap may also be applied.

Reduced data are written to rolling Parquet files. Selected waveform snippets are written to binary waveform files and indexed in SQLite. The archive is organized by year, month, day, and hour.

The program also supports live visualization through a preview pathway that is kept separate from the main acquisition loop.

Runtime controls include timeout handling, configurable pauses, and automatic rearming.

# 4 Installation and user setup

## 4.1 Adding a new user

If a new Linux user account must be created for instrument operation and should inherit the same graphical environment and theme configuration as an existing account, the following procedure may be used.

**Procedure: User configuration migration**

Once the new user profile is created with `sudo` access, execute the following script to synchronize the desktop environment and themes from `user1` to `user2`:

```
# Copy configuration and local data
sudo cp -ra /home/user1/.config /home/user2/
sudo cp -ra /home/user1/.local /home/user2/

# Copy theme and icon folders if they exist
sudo cp -ra /home/user1/.themes /home/user2/ 2>/dev/null
sudo cp -ra /home/user1/.icons /home/user2/ 2>/dev/null

# Ensure the new user owns their home directory
sudo chown -R user2:user2 /home/user2/
```

# 5 Operating procedure

## 5.1 General acquisition intent

The goal of CAPPY is to capture pulses reliably with sufficient pre-trigger and post-trigger coverage to preserve baseline and pulse-tail information.

The record window should be long enough to capture the full pulse, but short enough to avoid unnecessary storage.

## 5.2 Recommended startup procedure

1. Unlock the settings panel if necessary. The current password is `FermiAdmin1234`.
2. Run `Quick Config` to generate an initial trigger level and baseline configuration.
3. If needed, use the AlazarTech front-panel software to inspect the incoming signal directly.
4. During initial tuning, set `Save waveform every N` to 1.
5. Adjust pre-trigger and post-trigger samples until the full pulse is captured cleanly.
6. After tuning, increase `Save waveform every N` as needed for the run.
7. Monitor the live display, write status, and archive during acquisition.

# 6 Graphical user interface

## 6.1 Main acquisition GUI

Upon launch, the user is presented with the main acquisition interface shown in Figure 1.

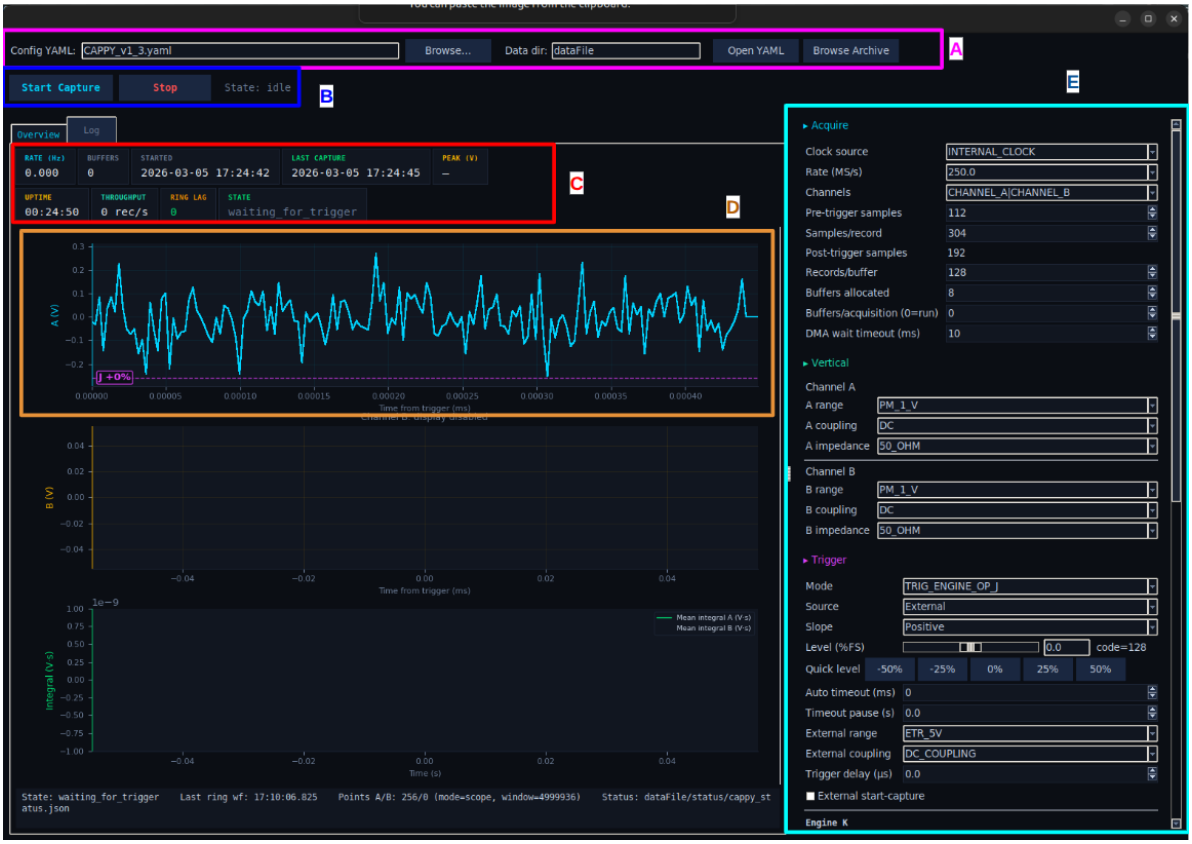


**Figure 1:** Main CAPPY graphical user interface.

**A. File and archive controls.** Selects the YAML file, data directory, and archive tools.

**B. Capture controls.** Starts or stops acquisition and displays the current software state.

**C. Overview metrics.** Displays rate, buffers, start time, last capture time, uptime, throughput, and related status fields.

**D. Live display.** Shows waveform preview and reduced observables.

**E. Configuration panel.** Contains acquisition, input, trigger, storage, runtime, and quick-configuration settings.

# 7 Configuration reference

The right-hand configuration panel controls acquisition, input configuration, triggering, runtime behavior, storage, and live display.

## 7.1 Acquisition settings

**Table 1:** Acquisition settings in the `Acquire` section.

| Setting | Description |
|---|---|
| Clock source | Selects the sampling clock source. |
| Rate (MS/s) | Sets the sampling rate. |
| Channels | Selects the active input channels. |
| Pre-trigger samples | Number of samples recorded before the trigger. |
| Samples/record | Total samples stored per record. |
| Post-trigger samples | Number of samples recorded after the trigger. |
| Records/buffer | Number of records grouped into each DMA buffer. |
| Buffers allocated | Number of DMA buffers reserved in host memory. |
| Buffers/acquisition (0=run) | Total buffers acquired before stopping; 0 means continuous operation. |
| DMA wait timeout (ms) | Maximum wait time for a DMA buffer to complete. |

## 7.2 Input settings

**Table 2:** Input settings in the `Vertical` section.

| Setting | Description |
|---|---|
| A range / B range | Sets the full-scale input range for Channel A or B. |
| A coupling / B coupling | Selects the input coupling mode. |
| A impedance / B impedance | Sets the input termination impedance. |

## 7.3 Trigger settings

**Table 3:** Trigger settings in the `Trigger` section.

| Setting | Description |
|---|---|
| Mode | Selects the trigger engine mode. |
| Source | Selects the trigger source. |

| Setting | Description |
|---|---|
| Slope | Selects rising- or falling-edge triggering. |
| Level (%FS) | Sets the trigger threshold as a percentage of full scale. |
| Quick level | Applies a preset trigger level. |
| Auto timeout (ms) | Sets the auto-trigger timeout interval. |
| Timeout pause (s) | Sets the pause after a timeout event. |
| External range | Sets the external trigger input range. |
| External coupling | Sets the external trigger coupling mode. |
| Trigger delay ($\mu$s) | Delays the trigger reference point. |
| External start-capture | Enables external start control. |
| Engine K source | Selects the secondary trigger source. |
| Engine K slope | Selects the secondary trigger slope. |
| Engine K level (%FS) | Sets the secondary trigger threshold. |

## 7.4 Runtime, storage, and live-display settings

**Table 4:** Runtime, storage, live-display, and quick-configuration settings.

| Setting | Description |
|---|---|
| Runtime profile | Loads a predefined runtime profile. |
| Rearm if no trigger (s) | Rearms acquisition after the specified no-trigger interval. |
| Rearm cooldown (s) | Minimum delay between rearm attempts. |
| Max rearms/hour | Maximum automatic rearms per hour. |
| Flush every records | Flushes reduced data after the specified record count. |
| Flush every seconds | Flushes reduced data after the specified time interval. |
| SQLite commit every snips | Commits snippet index data after the specified snippet count. |
| Session rotate (hours) | Starts a new session after the specified runtime. |
| Ring slots | Number of live-preview ring-buffer slots. |

| Setting | Description |
|---|---|
| Ring points | Number of points stored per preview update. |
| Live window points | Total points shown in the live display window. |
| Live window seconds | Time span represented by the live display window. |
| Max waveforms/tick | Maximum waveforms processed per GUI update. |
| Save waveform every N | Saves one waveform for every $N$ accepted events. |
| Live waveform every N buffers | Updates the live display every $N$ DMA buffers. |
| Saved waveforms/sec cap | Maximum waveform-save rate per second. |
| Show Channel B live waveform | Enables Channel B in the live display. |
| Preview mode | Selects the live preview mode. |
| Quick Config | Runs the automatic setup routine. |
| Save Controls To YAML | Saves current GUI settings to the YAML file. |
| Reload Controls From YAML | Reloads GUI settings from the YAML file. |
| Disk write status | Displays current archive write status. |

# 8 Archive browser

## 8.1 Purpose of CAPPY.ARCH

CAPPY.ARCH is the archive browser for the CAPPY acquisition system. It reads the saved archive and displays waveform snippets, reduced quantities, timestamps, and indexing metadata.

## 8.2 Archive layout

The main archive interface is shown in Figure 2. The left side contains navigation tables and the snippet list. The right side displays the selected waveform, integral trace, and metadata.

**Figure 2:** Main CAPPY.ARCH interface.

The principal navigation regions are:

- **Sessions:** Lists acquisition sessions.
- **Day:** Groups the archive by day.
- **Hour:** Narrows the selection by hour.
- **Minute:** Narrows the selection by minute.
- **Snips:** Lists saved waveform snippets in the selected interval.

## 8.3 How waveforms enter the archive

The program applies the configured waveform-selection policy before storage.

## 8.4 Meaning of the archive identifiers

The archive view exposes multiple indexing fields for each saved snippet.

**Table 5:** Meaning of identifier fields shown in the archive browser.

| Field | Meaning |
|---|---|
| Timestamp | Wall-clock time of the saved waveform snippet. |
| ID | Archive snippet identifier. |
| Buf | DMA buffer number. |
| Rec | Record index within the DMA buffer. |
| Global | Global acquisition record index. |

For example, if the archive displays `ID 589`, `Buf 180`, `Rec 38`, and `Global 23078`, then the displayed waveform is saved snippet 589, taken from DMA buffer 180, record 38, and global record 23078.

### 8.5 Metadata panel

The metadata panel beneath the plots typically includes:

- snippet number,
- timestamp,
- session identifier,
- DMA buffer number,
- record number,
- global record index,
- sampling rate,
- waveform length,
- integrated area,
- peak amplitude.

## 9 Waveform inspection tools

### 9.1 Standard waveform display

When a snippet is selected in CAPPY.ARCH, the right side of the interface displays two linked plots. The upper plot shows the waveform and the lower plot shows the corresponding integral or reduced-area trace.

## 9.2 Draggy Mode

Draggy Mode allows the user to define a time interval directly on the waveform plot. The same interval is then applied to the corresponding integral display.

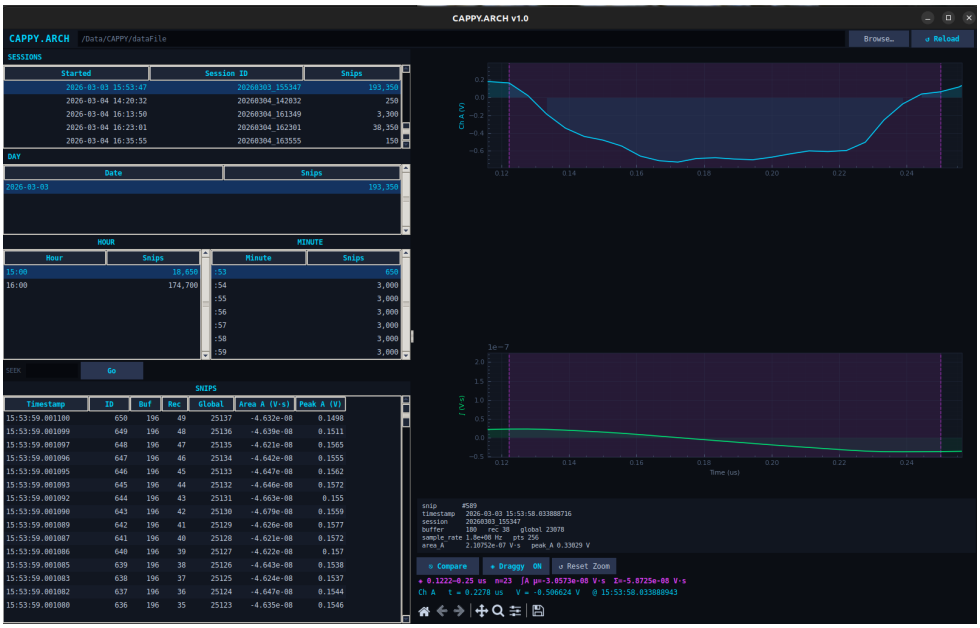


**Figure 3:** Draggy Mode in CAPPY.ARCH.

This mode is useful for restricting attention to a selected waveform region.

## 9.3 Compare mode

Compare mode opens the selected waveform in a separate comparison window.

**Figure 4:** Comparison window in CAPPY.ARCH.

The title bar of the comparison window shows the session and snippet identifier.

# 10 Interpretation guidelines

When working with archived waveforms:

1. A **saved snippet** is not necessarily every acquired event.
2. The **ID** field is the saved-snippet index in the archive.
3. The **Global** field gives the position in the full acquisition stream.
4. **Buf** and **Rec** give the event location within the DMA structure.
5. The combination of **session + ID + Global** is often the most precise identifier.

# 11 Author notes

This guide is intended as an operating reference for CAPPY and CAPPY.ARCH. C migration soon.

For troubleshooting or support: naianyza@hawaii.edu

Project repository: https://github.com/naiannyza-lang/CAPPY